Travis County Congressional Districts Analysis- A study of Possible Gerrymandering

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GISC 1391

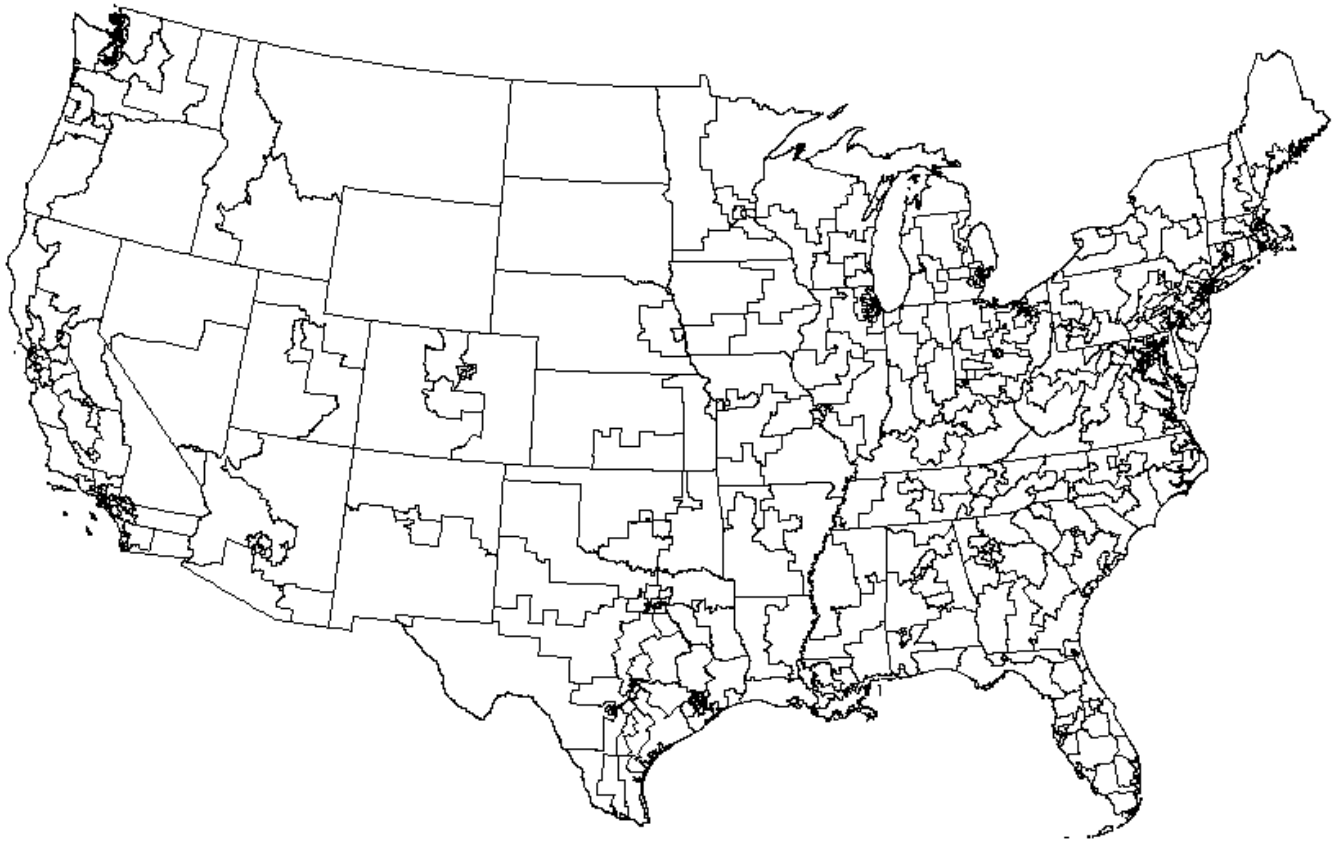
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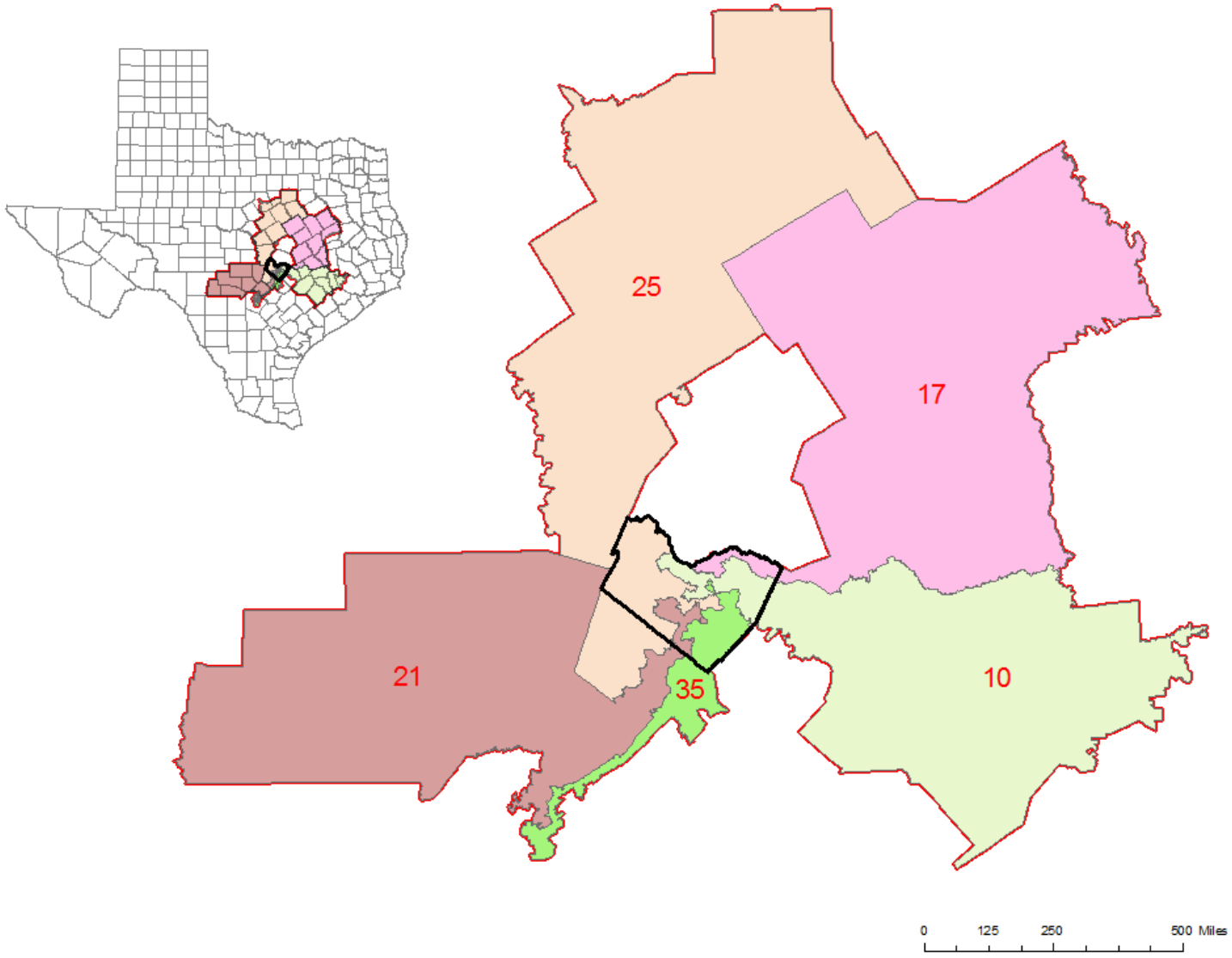
Background: Congressional Districts:

- House of Representatives
  - Capped at 435 members
- House seats are reapportioned after each decennial census with some states gaining and others losing seats.
- Each should closely be of equal population within the state.
- Any population difference greater than one percent between the most and least populated districts within the state will likely be unconstitutional.
- Note: VTD will be used for the term “Voting Tabulation Districts”

Background: US Congressional Districts
114th Congress January 2015-2017



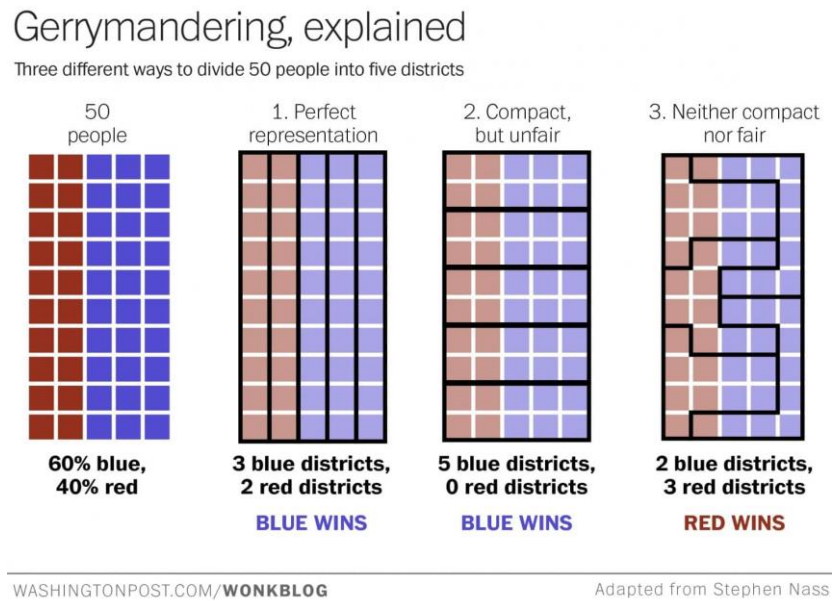
Background: US Congressional Districts Representing Travis County



10th (Michael McCaul-R)
17th (Bill Flores-R)
21st (Lamar Smith-R)
31st (John Carter-R)
35th (Lloyd Doggett-D)

Background: What is gerrymandering?

- Packing: pack as many of your opposing party's voters into as few of voting districts as you can.
- Cracking: Spread the remainder of these voters to as many other districts as you can.



Source: https://www.washingtonpost.com/news/wonk/wp/2015/03/01/this-is-the-best-explanation-of-gerrymandering-you-will-ever-see/?noredirect=on&utm_term=.b8c24a7ddec6

Research Questions

- Research Questions:
  - Is there a significant difference between the racial and political breakdown of each congressional district in question in Travis County with the portion of the district outside Travis County?
  - Do the shapes of the congressional districts representing Travis County violate any standards and considered gerrymandered?

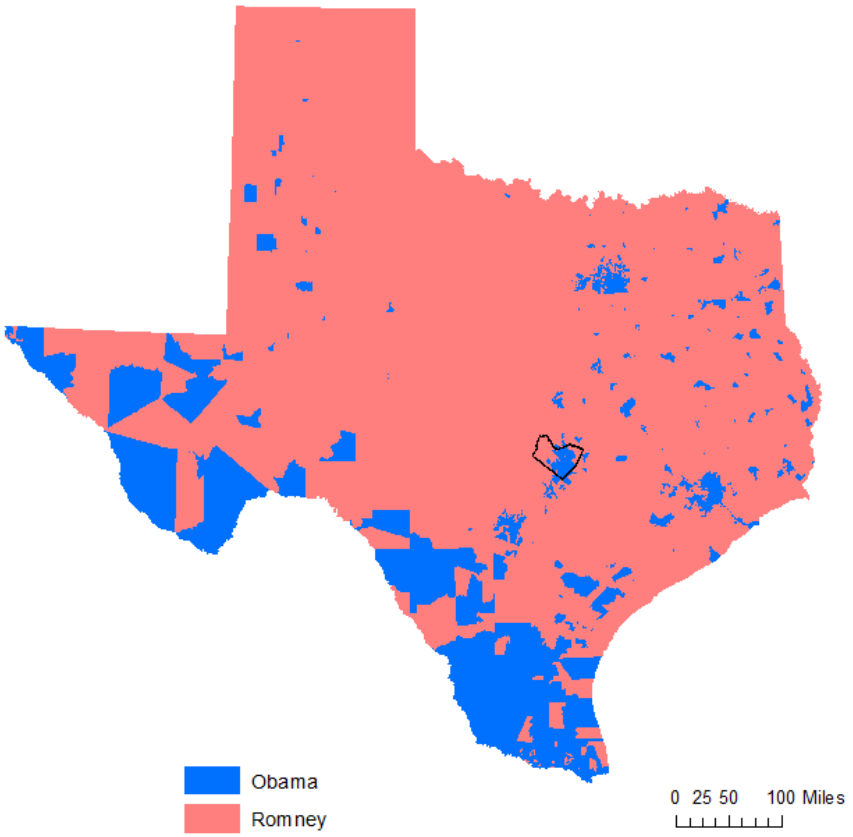
Methods

- VTDs Files
  - representations of voting precincts using census block geography. Counties can use whatever line they prefer to draw voting precinct lines, so a VTD would use the nearest census block boundary if a line is not in census geography.
- 2010 census population subtotals are already in the VTD Shapefile
- $circular\ index = perimeter \div (2 \times \pi \times \sqrt{(area \div \pi)})$
  - Circular Index of Circle = 1
  - Circular Index >1.67 Considered too stretched to be representive
- ArcMap 10.5.1
- File Geodatabase: Travis_County_Cong.gdb
- Personal Geodatabase: Gerrymandering.mdb
- Coordinate System: NAD1983 StatePlane Texas Central FIPS 4203 Feet
- Projection: Lambert_Conformal_Conic

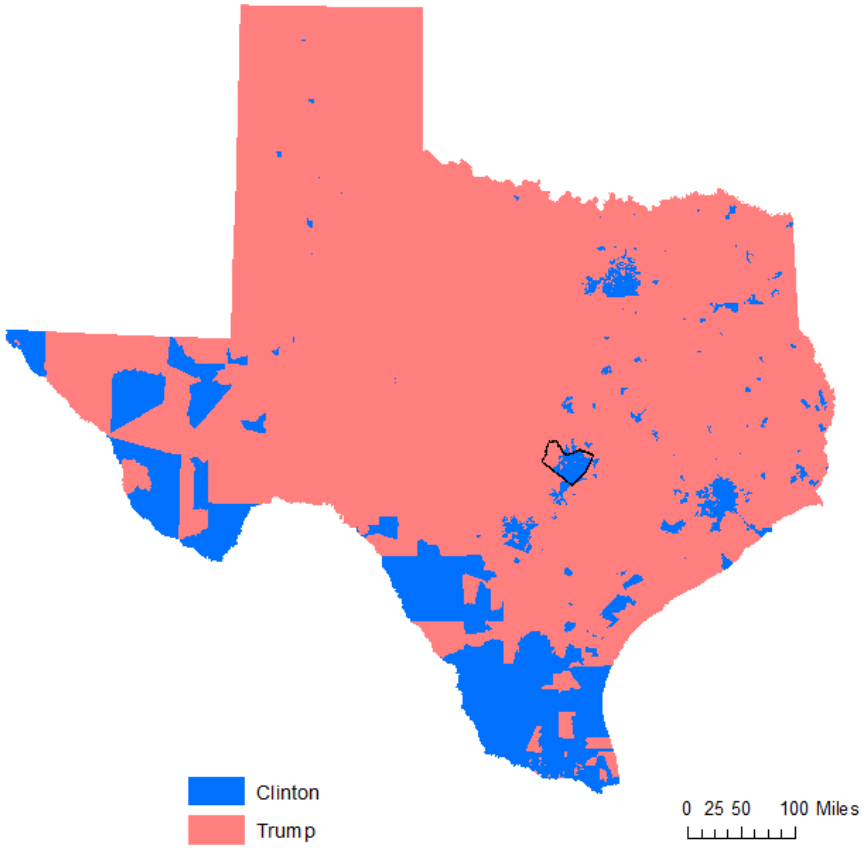
Comparing Presidential Elections

By Voting District

2012



2016

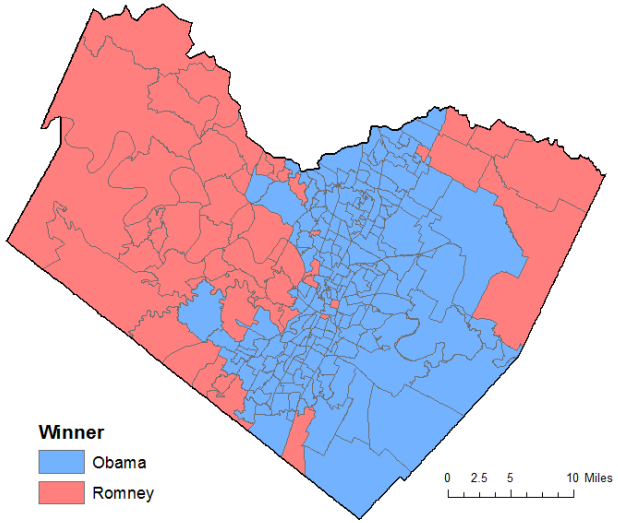


Travis County Election Results

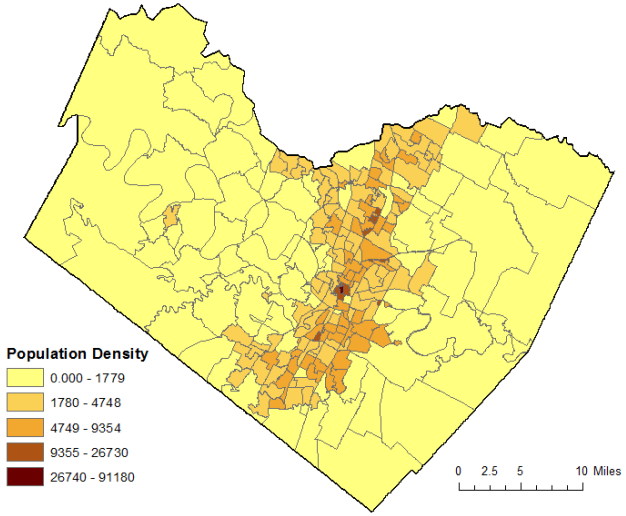
Year	Straight Ballot (%)	Presidential Election (%)
2012	Democratic: 61 Republican: 36	Obama: 60 Romney: 36
2016	Democratic: 66 Republican: 30	Clinton: 65 Trump: 30

Travis County Political Makeup

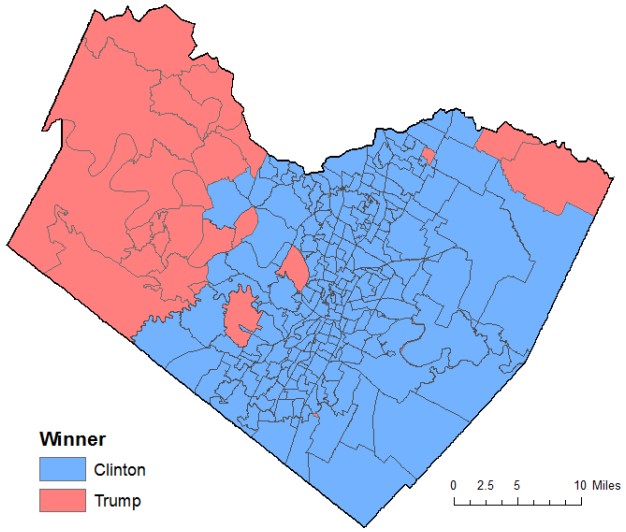
Travis County 2012 Presidential Election by VTD



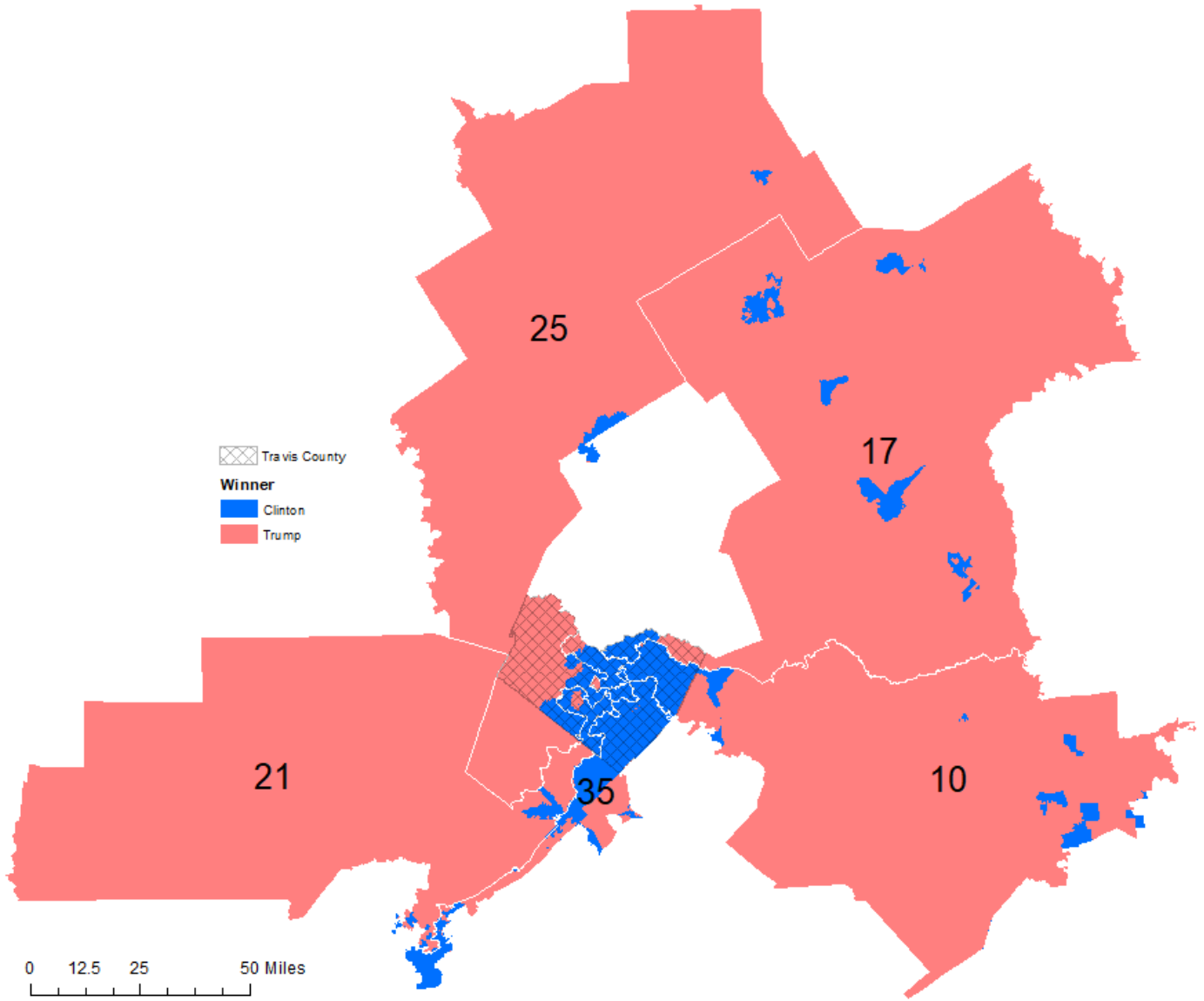
Travis County 2010 Population Density Per Square Mile By VTD



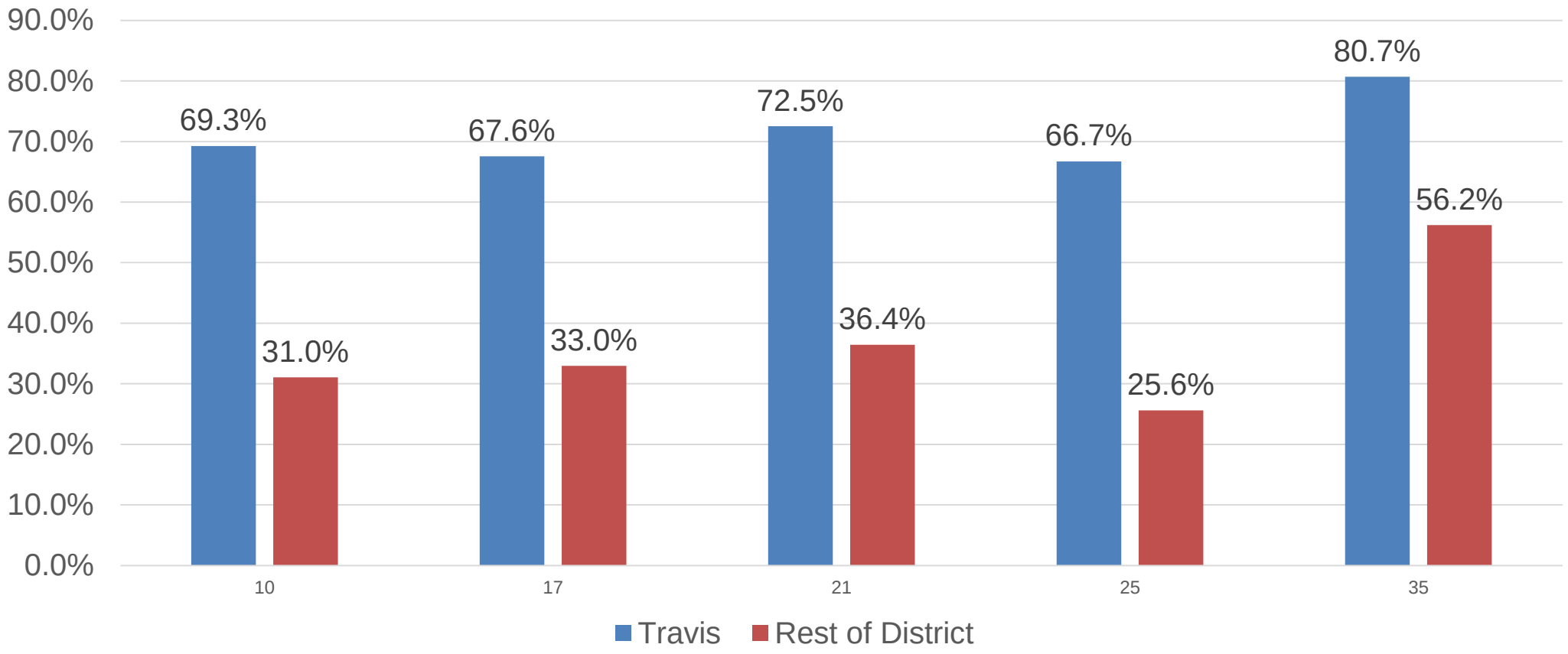
Travis County 2016 Presidential Election by VTD



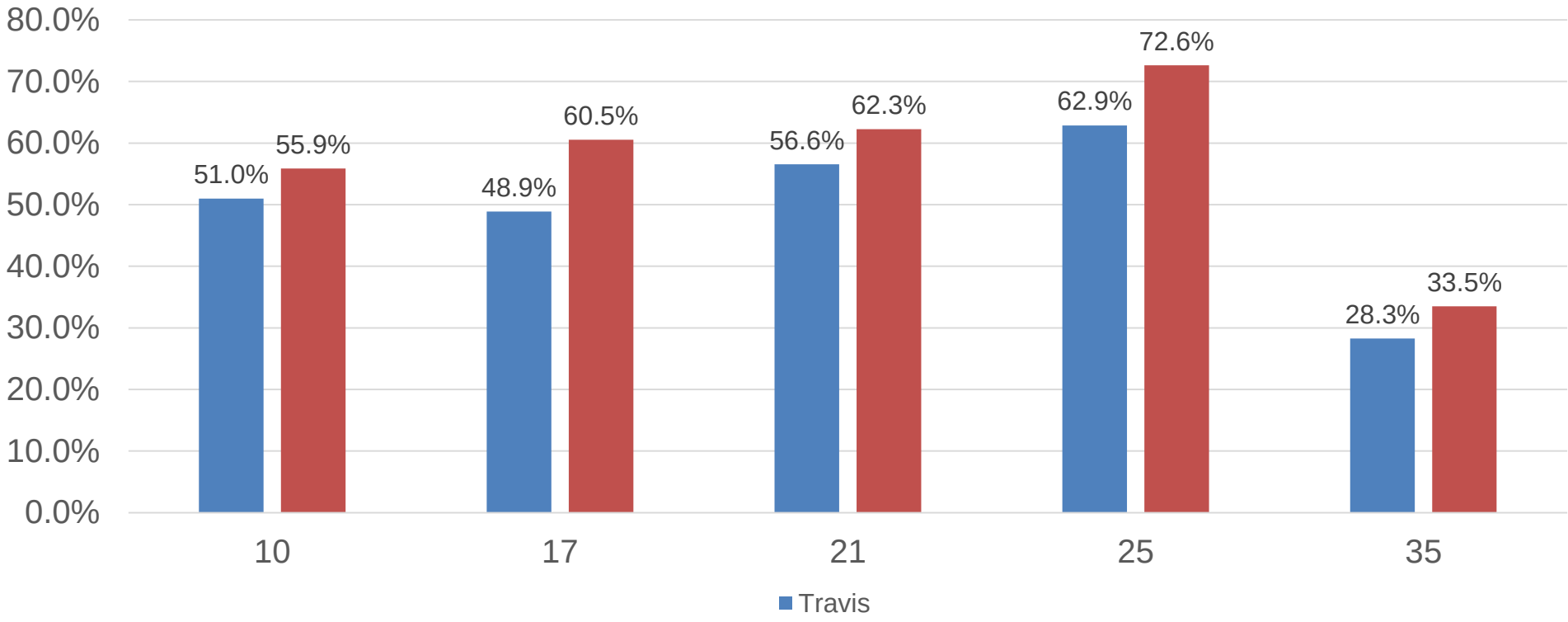
2016 Presidential Election Results



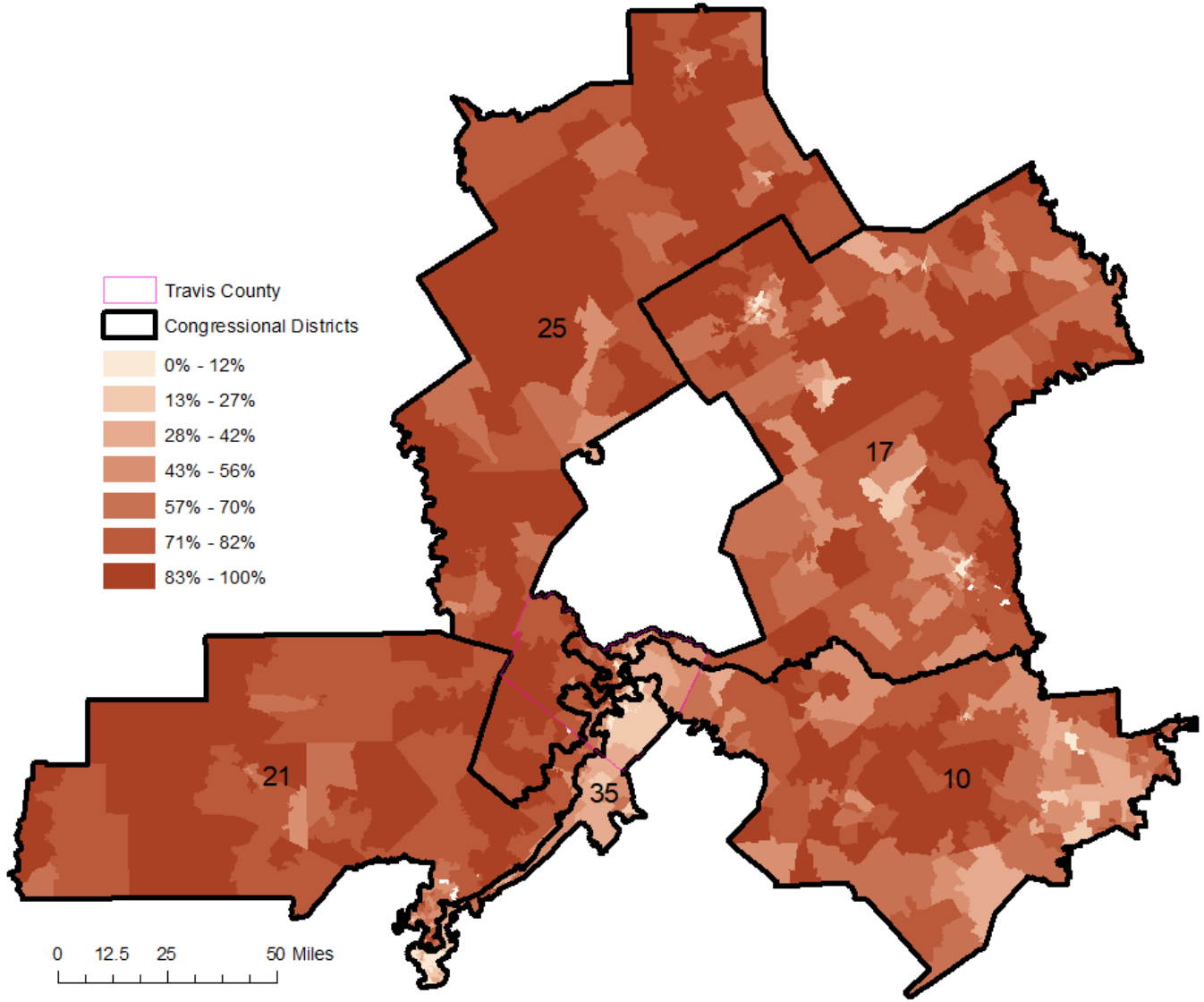
Percent for Clinton Travis County versus Rest of District



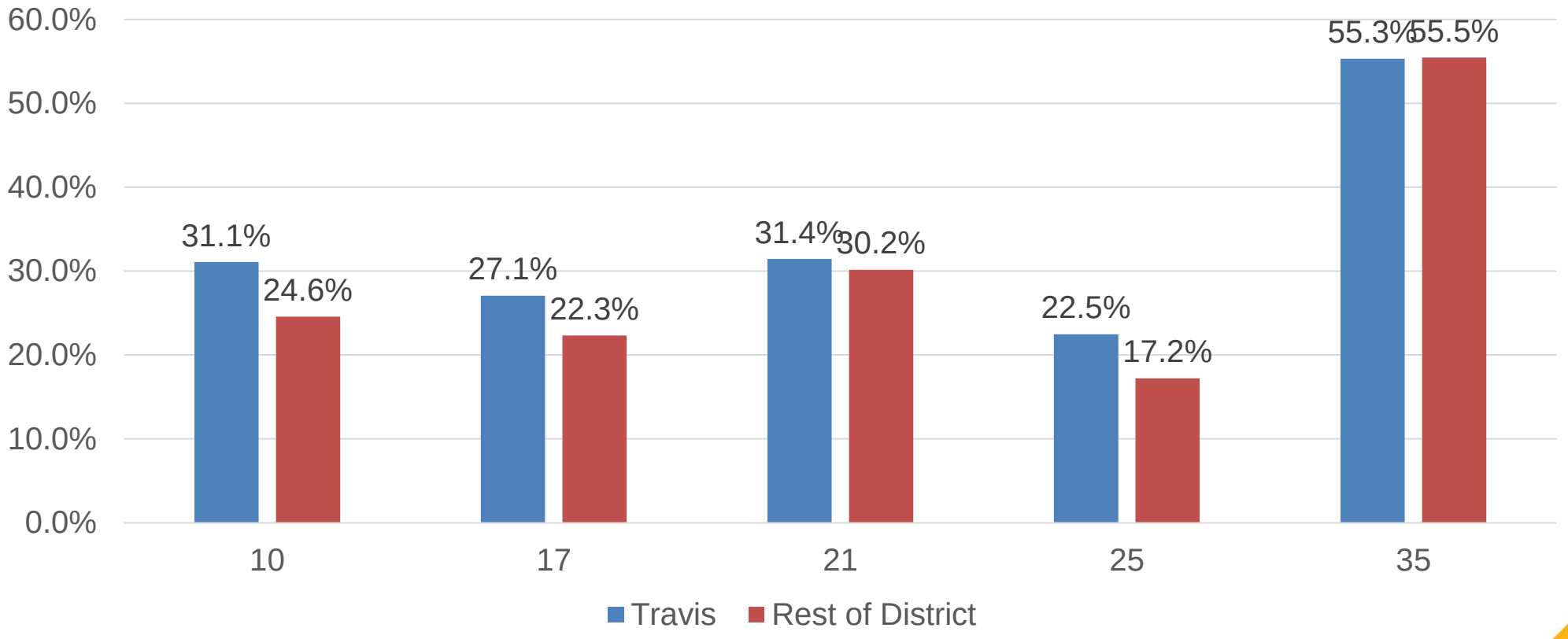
Percent White Travis County versus Combined Rest of District



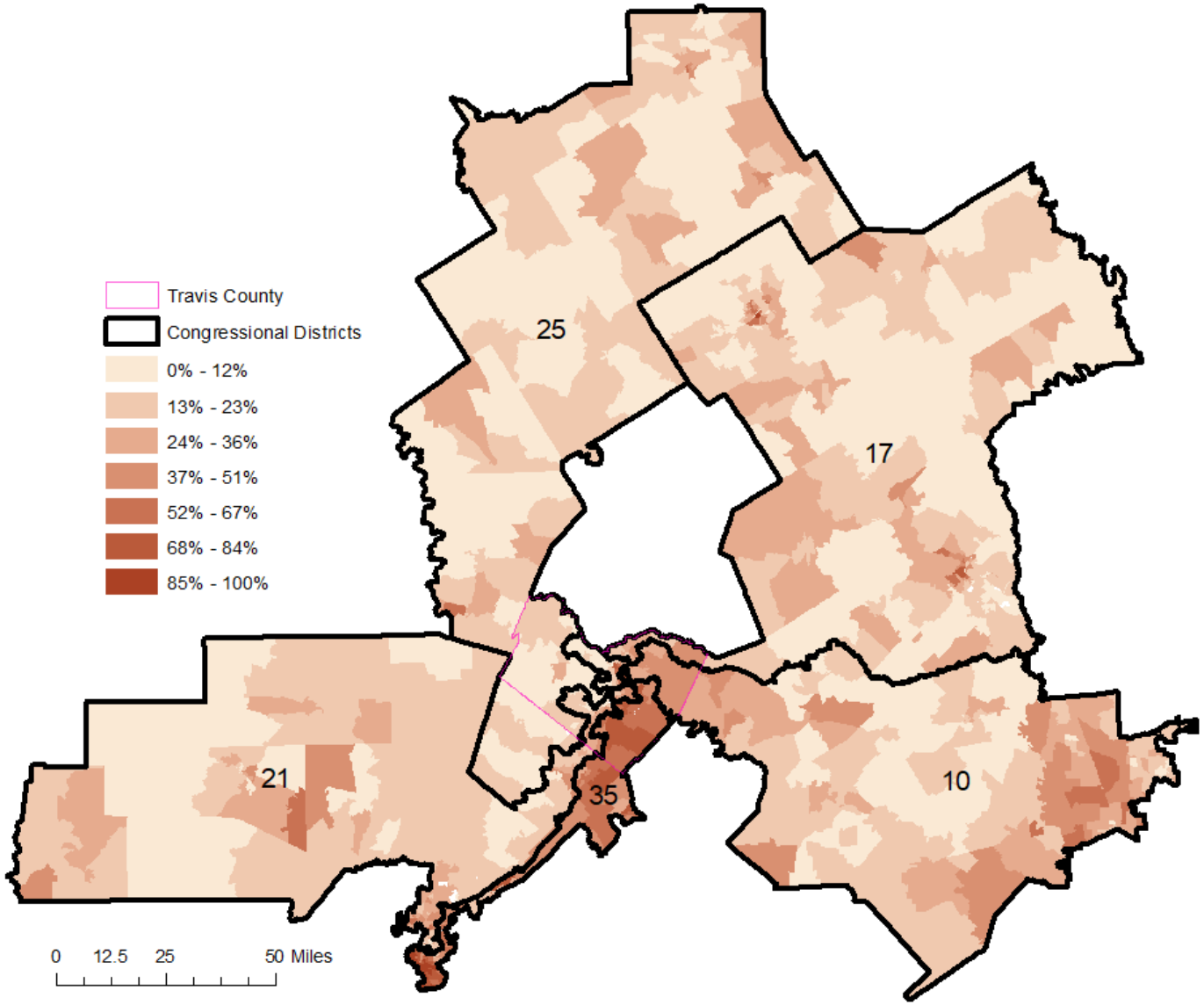
Percentage White Per VTD 2010 Census



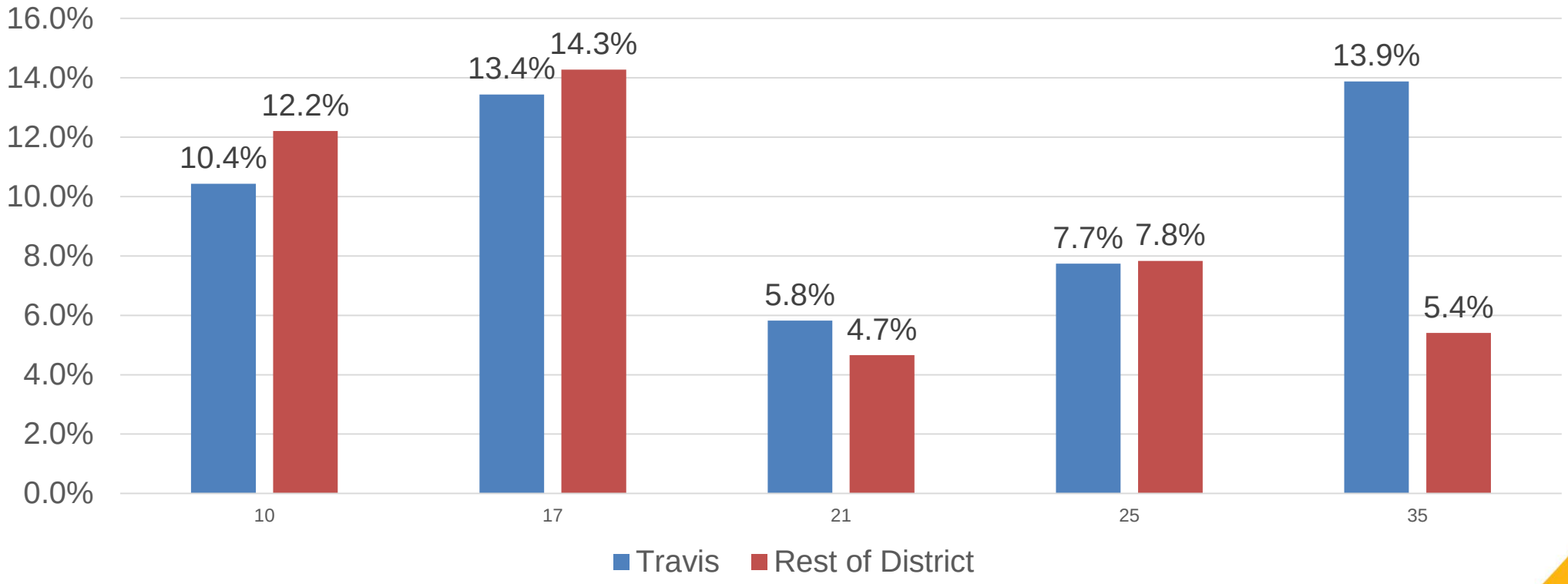
Percent Hispanic Travis County versus Combined Rest of District



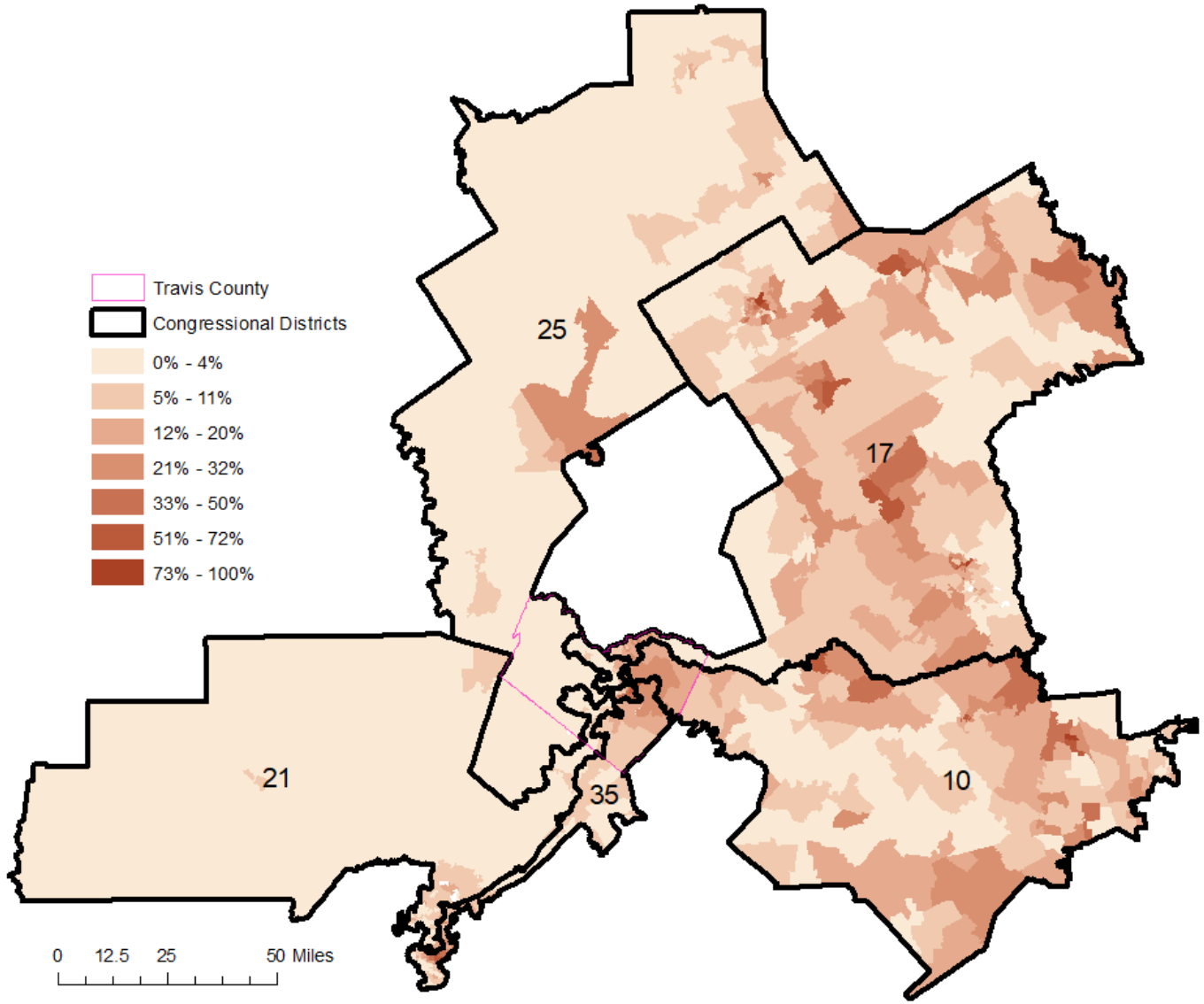
Percentage Hispanic Per VTD 2010 Census



Percent Black Travis County versus Combined Rest of District



Percentage Black Per VTD 2010 Census



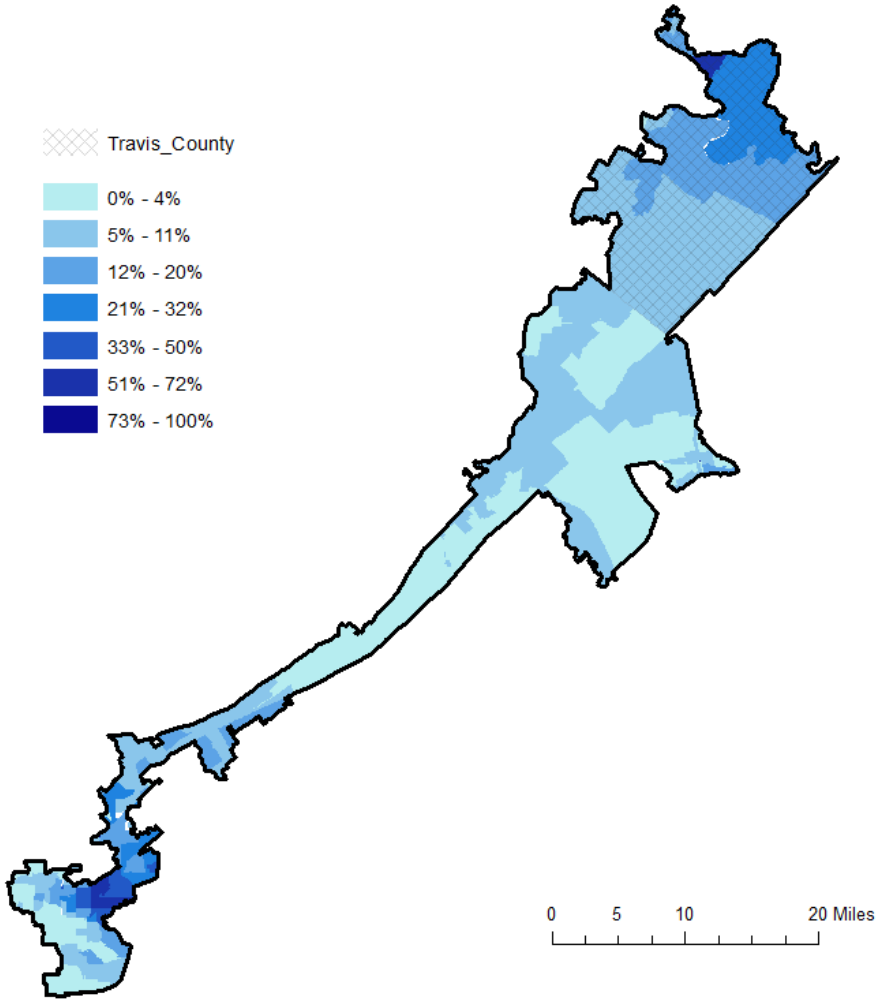
Circular Index of Congressional Shapes

District	Representation	Area (Sq. Mi)	Perimeter (Mi)	Circular Index
10	Michael McCaul-R	5143.7	632.3	2.49
17	Bill Flores-R	7768.8	693.3	2.21
21	Lamar Smith-R	5956.8	617.9	2.26
25	John Carter-R	7744.4	774.5	2.48
35	Lloyd Doggett-D	601.5	367.4	4.23

Summary

Dist	Rep	Circular Index	2016 Pres Election Percentage Clinton		Racial Representation Percentage					
			Travis	*Dist*	*Travis*	*Dist*	*Travis*	*Dist*	*Travis*	*Dist*
					White	*White*	*Hisp*	*Hisp*	*Black*	*Black*
10th	R	2.49	69.3	31.0	51.0	55.9	31.1	24.6	10.4	12.2
17th	R	2.21	67.6	33.0	48.9	60.5	27.1	22.3	13.4	14.3
21st	R	2.26	72.5	36.4	56.6	62.3	31.4	30.2	5.8	4.7
31st	R	2.48	66.7	25.6	62.9	72.6	22.5	17.2	7.7	7.8
35th	D	4.23	80.7	56.2	28.3	33.5	55.3	55.5	13.9	5.4

US Congressional District 35 Percentage Black Per 2010 Census VTD



Conclusions

- Research Questions / Answers:
  - Is there a significant difference between the racial and political breakdown of each congressional district in question in Travis County with the portion of the district outside Travis County?
    - There is indeed a significant difference in how voters within Travis county vote as compared to the rest of the Congressional District.
    - No significant differences between the racial breakdown between the population living within Travis County as compared with the rest of the Congressional District with the exception of District 35, where the percentage of the black population inside Travis County is over twice of that outside Travis County.

Conclusions Cont.

- Research Questions / Answers Cont.:
  - Do the shapes of the congressional districts representing Travis County violate any standards and considered gerrymandered?
    - If we go by 1.67 as the maximum value allowed for a Congressional District, then all of the Congressional Districts violate the acceptable roundness criteria.
    - Congressional District 35 is especially misshaped for representation.
    - It is no surprise that Congressional District 35 is under scrutiny and we are currently waiting a court ruling to see whether it needs to be redrawn or not.

Acknowledgements

- Sally Holl (ACC)
  - Setting up the personal database
- Charles Eckstein (Texas Legislative Council)
  - VTD shapefiles
  - 2010 census population data combined with election

References

Title	Link
Texas Legislative Council FTD files	ftp://ftpgis1.tlc.state.tx.us/2011_Redistricting_Data/VTDs/Geography/
Texas Legislative Council Election Results	ftp://ftpgis1.tlc.state.tx.us/elections/
The Geeks Who Put a Stop to Pennsylvania's Partisan Gerrymandering	https://www.wired.com/story/pennsylvania-partisan-gerrymandering-experts/
US Census Bureau: Congressional District Shapefiles	https://www.census.gov/geo/maps-data/data/cbf/cbf_cds.html
Analysis with QGIS: Congressional Gerrymandering?	https://www2.hawaii.edu/~matt/104/Exercises/qgis-analysis-gerrymander.html

Contact Me For Questions/Comments

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